Heapsort exercises

06-E01 Heapify trace

Difficulty: Foundational.

A=[4,9,8,3,7,6,5] has max-heap child subtrees. Heapify the root, listing every swap and the stopping condition.

06-E02 Node heights and linear building

Difficulty: Advanced.

For the perfect binary tree n=15, count nodes at each height h and compute the rough cost sum count(h)・h. Compare with O(n).

06-E03 Priority update

Difficulty: Advanced.

In max-heap A=[20,15,18,7,12,9,17,2], increase position 8 from 2 to 16. Trace increase-key and identify the necessary condition for each swap.

06-E04 An incorrect parent-index validator

Difficulty: Advanced.

A 1-based validator checks A[ceil(c/2)]≥A[c] for every c=2..n. Test it on A=[10,1,9]. Give the correct condition and its 0-based counterpart.

06-E05 Stability with equal keys

Difficulty: Advanced.

Compare records (2,a),(2,b),(1,c) only by key. Trace identities through heapsort with heapify that does not exchange equal keys. Is it stable?

06-E06 Two repair directions after deletion

Difficulty: Research.

Given max-heap A=[50,20,45,10,15,40,44,5,9,12,14,35,39,42,43], start independently from this original array for each deletion. Delete position 4, then independently position 1, filling with the last element. List swaps and final arrays. Prove how to choose upward or downward repair.

Work independently before opening the separate solutions PDF.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Heapsort exercises (continued)

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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